

Tyler Drake

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior software engineer who builds identity and risk platforms that are dependable under operational and regulatory pressure. At Charles Schwab, built the **Enhanced Client Verification platform** across robust identity verification, fraud-risk detection and mitigation, AML mitigation, account-linking and verification orchestration, and sanctioned-watchlist screening. Experienced with Java 17, Spring Boot, Kafka, Avro, SQL, secure delivery, observability, AIDLC delivery practices, and agentic developer tooling.

TECHNICAL SKILLS

Languages: Java 17, Python, SQL, Bash, JavaScript

Application: Spring Boot 3, REST, WebFlux, Maven, JUnit, Cucumber BDD, HTML, CSS

Messaging and data: Apache Kafka, Avro, RabbitMQ, MongoDB, SQL Server

Delivery and operations: GitHub Actions CD, Bamboo, Pivotal Cloud Foundry, Splunk logging and telemetry, production monitoring and alerting, JIRA, Confluence

Platform and tooling: SQLite/SQLCipher, PowerShell, pytest, MCP server development

Security and quality: Quality Assurance and quality engineering, trusted committer/code review, Veracode, Black Duck, SonarQube, certification, secure software delivery

Practices: Agile, event-driven design, CI/CD, resiliency, Blue/Green deployments, big-bang deployments, AIDLC practices for AI-assisted delivery workflows

Migration and storage: legacy-data migration, modern Hotlist migration, S3-compatible object storage

PROFESSIONAL EXPERIENCE

Charles Schwab - Senior Software Engineer

06/2018 - Present

Denver, Colorado, United States

- Own responsibilities across Quality Assurance, Development and Engineering, DevOps CI/CD pipeline tooling, Trusted Committer code reviews, production monitoring and alerting with Splunk logging and telemetry, resiliency, certification, Blue/Green and big-bang deployments, and leading AI adoption across the organization; create architecture documentation, present SAD materials to Local Governance committees, and own the Service Registry for platform discoverability.
- Built the **Enhanced Client Verification platform** from the first line of code: event-driven services for robust identity verification, fraud-risk detection and mitigation, AML mitigation, account linking and verification orchestration, and sanctioned-watchlist screening.
- Currently leading a legacy-data migration to a modern **Hotlist** for bad-actors screening and monitoring.
- Migrated SPOS object storage to **S3-compatible buckets** supporting multiple integrations.
- Led the versioned **V1 and V3 events** migration by dual-publishing Kafka/Avro events and moving orchestrators and consumers to the V3 contract without a service-wide interruption.
- Implemented IAM change events, leader election for single-publisher behavior, and lifecycle events including VerificationInFlight and AccountReactivated.
- Designed the FraudRisk flow and applied OA-Lock on DENY-FRAUD identity-verification decisions to strengthen account-restriction handling.
- Integrated IDVerse identity verification; tuned the GIACT decision matrix for RToo responses and authored the bank-risk verification design; delivered PAV eligibility changes and rescreening across ECV services, Recon, and API predecessors.
- Established GitHub Actions CD for ECV service and the actions orchestrator, migrated Hotlist, PIV, drainer, and Recon configuration repositories to GitHub, and moved rollout controls to LaunchDarkly flags.
- Upgraded services to Java 17 and Resilience4j, diagnosed memory-crash behavior in ECV-API under load, created the BatchMaster Spring Boot skeleton, and added per-record deduplication across the API and generated client.
- Led the async PIV cutover and retired the RabbitMQ events interface after confirming no live traffic; delivered MRA events and maintained the AML data export for audit workflows.
- Developed a read-only LaunchDarkly MCP server, a Prod Readiness Composer, and a clean-coding agent skill; applied AIDLC practices to these AI-assisted delivery workflows, including MCP server development and agent harness design grounded in ECV delivery work.
- Led Agile planning and cross-functional delivery while using Veracode, Black Duck, and SonarQube practices to support secure, maintainable releases.

IBM - Developer

03/2016 - 06/2018

Monroe, Louisiana

- Built automation around REST and SOAP web-service development, including generated project artifacts, schemas, source files, and developer workspace setup.
- Analyzed service dependencies and addressed web-service timeout issues through data-driven investigation.
- Documented and presented high-level architecture for DVAR-OM web services, including layers, integrations, schemas, workflows, and environment configuration.
- Served as technical lead for web applications supporting customer order flow and telecommunications service provisioning.

Great-West Financial - Developer

04/2015 - 11/2015

Greenwood Village, Colorado

- Developed database-driven retirement-letter front ends with Thunderhead and integrated Business Objects; completed more than 75 independent projects.

EDUCATION

Colorado State University - Bachelor's degree in Computer Science, 01/2006 - 12/2009